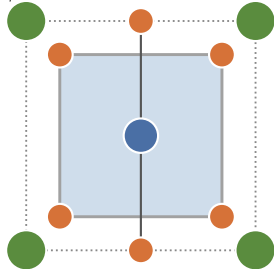


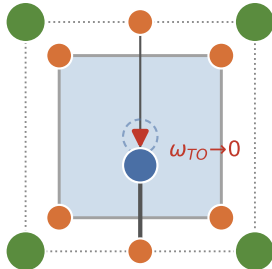
ABO_3 perovskite — three physics mechanisms encoded in PCNN

(a) Ideal cubic ABO_3
 ϵ_r^{CM} Clausius-Mossotti baseline



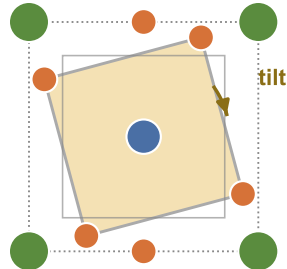
Polarizability of static lattice

(b) Soft-mode displacement
 δ_{LST} Lyddane-Sachs-Teller enhancement



$\epsilon_r \propto 1/\omega_{\text{TO}}^2$ · B-site off-centering

(c) Octahedral tilt
 δ_{tilt} Goldschmidt/Reaney suppression



$t = (r_A + r_O)/[\sqrt{2} (r_B + r_O)] < 1$ · reduces ϵ_r



A-site (e.g. Ba, Sr, Ca, La, Pb)



B-site (e.g. Ti, Zr, Nb, Mg)



Oxygen